# Do Now · retrieval
## Boolean algebra rules

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Normalise $0.0110 \times 2^4$ using a two’s-complement mantissa. Give the new mantissa and exponent.

### Q2

A sorted array is [3, 8, 12, 19, 27, 35, 42] . When searching for 27 , which indexes are checked, in order?

### Q3

Show the array after one full left-to-right pass of ascending bubble sort on [7, 3, 5, 2] .

### Q4

An alarm sounds if the door is open AND the system is armed, OR if panic is pressed. Write Q using A, B and C.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

De Morgan — CODE:
¬(A ∧ B) → ¬A ∨ ¬B
¬(A ∨ B) → ¬A ∧ ¬B — The operator inside the brackets swaps as each term is negated.

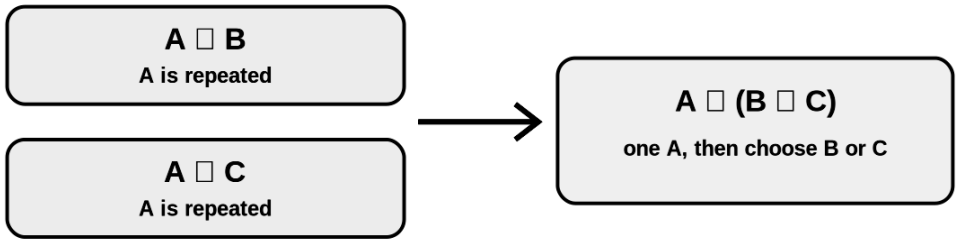
## Core idea 2

Double negation — CODE:
¬(¬B) → B — Two consecutive NOT operations cancel.

## Core idea 3

Distributive law — CODE:
(A ∧ B) ∨ (A ∧ C)
= A ∧ (B ∨ C) — A appears in both groups. Write it once outside the brackets, then join B and C with OR.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Core idea 4

Association + commutation — CODE:
A ∨ (B ∨ C)
= (A ∨ B) ∨ C

A ∧ B
= B ∧ A — These allow regrouping and reordering before other simplification.

## Vocabulary · use the correct technical terms

De Morgan — Rules for negating AND/OR groups.

distribution — Expanding/factoring Boolean expressions.

association — Regrouping repeated same operations.

commutation — Changing order.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Simplify ¬(A ∧ B)

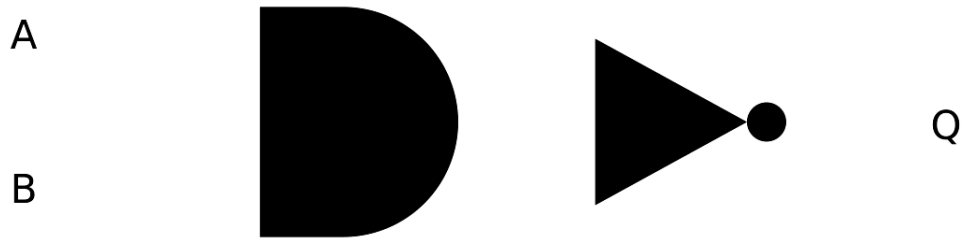
## Follow the worked method

1. Outer NOT over AND — CODE:
Use De Morgan's Law.

2. AND becomes OR — CODE:
¬A ∨ ¬B

3. Equivalent behaviour — CODE:
Both are false only when A=1 and B=1.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Complete the same expression
CODE:
¬A ∨ ¬B
= ¬(? ? ?)

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Simplify Boolean expressions

State the simplified form of each expression:

(i) ¬¬A (ii) ¬A ∧ ¬B (iii) ¬(¬A ∧ ¬B)

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

From a gate circuit to an expression
Two OR gates take inputs A, B and C, D respectively. Their outputs feed a third OR gate. Write the complete Boolean expression for the circuit output.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

De Morgan's Laws swap AND and OR while negating each term.

## Use key vocabulary from memory

1. Explain De Morgan in your own words.

2. Explain distribution in your own words.

3. Explain association in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.